Sensor signal processing

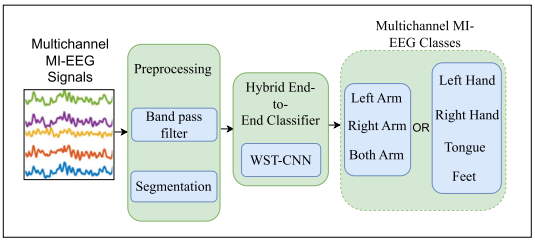
Classification of Motor Imagery Tasks Using EEG Based on Wavelet Scattering Transform and Convolutional Neural Network

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Abstract—Electroencephalogram (EEG) signal classification is of utmost importance in brain-computer interface (BCI) systems. However, the inherent complex properties of EEG signals pose a challenge in their analysis and modeling. This letter proposes a novel approach of integrating wavelet scattering transform (WST) with convolutional neural network (CNN) for classifying motor imagery (MI) via EEG signals (referred as WST-CNN), capable of extracting distinctive characteristics in signals even when the data is limited. In this architecture, the first layer is nontrainable WST features with fixed initializations in WST-CNN. Furthermore, WSTs are robust to local perturbations in data, especially in the form of translation invariance, and resilient to deformations, thereby enhancing the network's reliability. The performance of the proposed idea is evaluated on the DBCIE dataset for three different scenarios: left-arm (LA) movement, right-arm (RA) movement, and simultaneous movement of both arms (BA). The BCI Competition IV-2a dataset was also employed to validate the proposed concept across four distinct MI tasks, such as movements in: left-hand (LH), right-hand (RH), feet (FT), and tongue (T). The classifications' performance was evaluated in terms of accuracy (η), sensitivity (S_e), specificity (S_p), and weighted F1-score, which reached up to 92.72%, 92.72%, 97.57%, and 92.75% for classifying LH, RH, FT, and T on the BCI Competition IV-2a dataset and 89.19%, 89.19%, 94.60%, and 89.33% for classifying LA, RA, and BA, on the DBCIE dataset, respectively.



Index Terms—Sensor signal processing, brain-computer interface (BCI), convolutional neural network (CNN), DBCIE, electroencephalogram (EEG), motor imagery (MI), wavelet scattering transform (WST).

I. INTRODUCTION

Motor imagery (MI) classification tasks using electroencephalograms (EEGs) are critical to the operation of numerous brain-computer interfaces (BCI). Various BCI-based signal processing techniques based on EEG data such as clustering sparse swarm decomposition [1], multiscale domain gradient boosting models [2], and logistics regression with tangent space learning [3] have been proposed. In another attempt [4], authors tried to translate brain activities into machine commands for controlling external electronic devices by making a link between BCI and subsets of the EEG signal, using a data summarization technique from computational geometry for handling streaming data. Such devices are crucial in practical applications, such as successful neuro-motor rehabilitation procedures [5], precise control of robotic systems [6], stroke recovery [5], and virtual reality games [7]. Further, in the direction of features extraction from EEG data, several works in literature [8], [9], [10], [11] have been reported in the context of MI classification. Common approaches for feature extraction include wavelet transform (WT) [8], common spatial patterns [9], principal component analysis [10], and empirical mode decomposition [11]. Such traditional features are typically handcrafted and call for a certain degree of expertise. Furthermore, these approaches may struggle to capture the nonlinear and complex nature of EEG signals, resulting in poor performance in MI task classification. Hence, the autonomous extraction of distinctive features from EEG signals becomes a critical requirement.

Advancements in deep learning (DL) has facilitated easy extraction of discriminative features from nonlinear EEG signals, potentially addressing the aforementioned issues. However, DL techniques generally have a large number of trainable parameters, demand large datasets, and take very long to converge. These limitations make them impractical in real-time BCI applications.

Hence, in this work, we propose to extract highly discriminative features through a wavelet scattering transform (WST) layer, which is integrated with a convolutional neural network (CNN) model as the first layer [12]. The primary role of a CNN is feature extraction, which can be computationally demanding when using multiple convolutional layers. Our supposition states that the features extracted by WST reduce the need for several CNN layers, thereby easing the computational load. In addition, the WST features exhibit translational invariance, capturing multiscale oscillations, handling nonlinearity, nonstationary, and are resilient to signal deformations [12]. Such features allow the model to be trained better and converge faster. To the best of the authors' knowledge, such a WST-based CNN technique has never been explored for the purpose of classification of MI tasks using EEG signals. To that end, the contributions of the proposed work are as follows.

1) We introduce a novel WST-based approach to extract distinctive features from EEG signals, incorporating hierarchical and phase information, while ensuring invariance to translation and minimal deformation.

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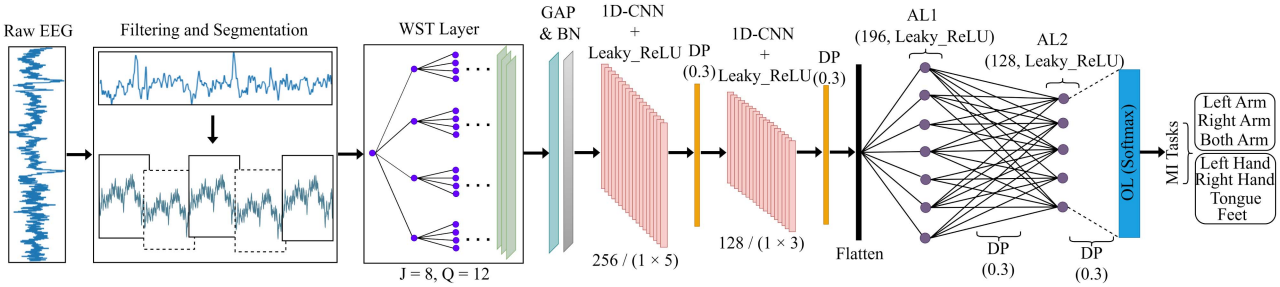


Fig. 1. Proposed WST-CNN network architecture for multiclass MI task classification. (Note: GAP ← global average pooling, BN ← batch normalization, DP ← dropout).

2) We integrate the WST with CNN to enhance feature representation, allowing better classification of MI tasks.
3) Besides the usual experiments, we also study the classification performance for multiple classes simultaneously, which has surprisingly never been reported in the literature before. Such an evaluation is critical to assess the model performance when several overlapping neuro-motor signals might be firing simultaneously (e.g., seizures).

Two separate datasets—BCI Competition IV-2a [13] and database for brain-computer interface using electrophysiological signal (DBCIE) [14]—were used for the experiments.

The rest of this letter is organized as follows. Section II discusses the state-of-the-art (SOTA) works. Section III presents the proposed approach, while Section IV details the experimental methodology. Section V discusses the results. Finally, Section VI concludes this letter.

II. PRIOR ART

Recent advancements in DL algorithms allow efficient processing of EEG signals for extracting discriminating features [15], especially using CNNs [16], [17]. The authors in [16] proposed a method called multifrequency band fusion (MFBF) for CNNs, which filters the input signal into different frequency bands, processes each with separate CNN models, and concatenates the outputs for the final classification of MI tasks. In [17], the authors proposed a method to convert 2-D time-series data to 3-D, extract features across time, space, and frequency domains, and then fuse these features for final classification. The authors of [18] proposed ATCNet, an attention-based convolutional network designed to enhance EEG-based MI classification by employing various techniques to improve performance while using a minimal number of parameters. In [19], the authors presented a DL-based approach to enhance recognition rates and attain balanced classification accuracy of MI tasks among different subjects through end-to-end learning. However, despite their success, DL approaches are typically computationally expensive, demanding large datasets, and converge slowly. To overcome the issues of DL techniques mentioned above, we propose a novel WST-CNN approach for classifying MI tasks.

III. PROPOSED WST-CNN NETWORK

This section discusses a hybrid end-to-end model, which combines the WST with a CNN. WSTs exhibit invariance to local perturbations, making the network significantly more reliable and efficient. In addition, it possesses the ability to extract valuable characteristics even when its kernels are fixed and can enhance performance on classification tasks with very limited data by providing useful, invariant, and generalized characteristics. It includes signal convolution, modulus nonlinearity, and averaging with a low pass filter (ζ). The wavelet $\psi(t)$ is a band-pass filter with center frequency λ normalized to 1, and a wavelet filter bank is defined as $\psi_\lambda(t) = \lambda\psi(\lambda t)$, where $\lambda = 2^{\frac{k}{Q}}$, for $k \in \mathbb{Z}$ and Q is the number of wavelets per octave. The bandwidth of $\psi(t)$ is approximately $\frac{1}{Q}$, resulting in a filter bank of band-pass filters centered at λ with a frequency bandwidth of $\frac{\lambda}{Q}$ [12]. The zero-order scattering coefficient, $S_0x(t) = x * \zeta(t)$, captures the signal's local translation invariance by convolving the original signal x with the scale function. However, this coefficient does not contain meaningful information for classification tasks, so we exclude it. First-order scattering coefficients are obtained as $S_1x(t) = |x * \psi_{\lambda_1}| * \zeta(t)$, which capture the local frequency content and significant features of the signal, resulting in representation that is invariant to translation. To maintain stability in the presence of time-warping deformations, the second-order wavelet modulus transformation is applied after the first WT and follows the Lipschitz deformation stability condition [12]. Scattering coefficients for mth order (i.e., $m \geq 2$) are obtained as, $S_{mx}(t, \lambda_1, \ldots, \lambda_m) = |||x * \psi_{\lambda_1}| * \ldots|\psi_{\lambda_m}| * \zeta(t)$.

Table 1. Specific Parameters of WST-CNN Network

Layer	Filter/Nodes	Kernel Size	No. of Parameters
Scattering1D	-	-	-
GAP1D + BN	-	-	1344
Conv1D + LeakyReLU	256	(1,5), stride = 1	1536
BN + MP1D + DP	-	DP = 0.3	1024
Conv1D + LeakyReLU	128	(1,3), stride = 1	98432
BN + MP1D + DP	-	DP = 0.3	512
Dense 1 + DP	196	DP = 0.3	2057412
Dense 2 + DP	128	DP = 0.3	25216
Dense 3 + Softmax	N	-	516

Note: GAP1D ← Global Average Pooling 1D, BN ← Batch Normalization, MP1D ← Max Pooling 1D, DP ← Dropout, N ← No. of Classes: N = 3 for DBCIE [14], N = 4 for BCI Competition IV-2a [13].

Fig. 1, depicts a hybrid end-to-end methodology of incorporating a nontrainable WST layer before a 1-D-CNN for the classification of MI task. The incorporation of the WST of order two ($m = 2$) is facilitated through the utilization of *kymatio* 1-D wavelets [20]. After WST, a 1-D global average pooling layer is employed, followed by a batch normalization (BN) layer. The output of the BN layer is subsequently fed into a 1-D convolutional block comprising two convolutional layers and leaky rectified linear unit (leakyReLU) nonlinear activation function. This combination leverages the strengths of both WST and CNN in extracting significant features of time-frequency (TF) representations of EEG data. WST finds important TF patterns and invariances in EEG data, resulting in a rich feature set that CNNs can further improve by adding convolutional layers. Furthermore, the WST layer, which acts as a feature extractor for the succeeding convolutional layers in the proposed network, helping to reduce the computational load on CNN (requiring few convolution layers). Moreover, WST-CNN can be trained on limited data samples without experiencing overfitting. Such advantage is not possible using the simpler and conventional multilayer perceptions (MLPs) and long short-term memory networks (LSTMs) [21]. MLPs struggle with high-dimensional data, and LSTMs are better suited for sequence processing but less effective at capturing spatial patterns within TF domains. In this architecture, only two convolutional layers were required to reduce the computational complexity. Following the convolution block, two affine layers are introduced for classification, with dropouts applied after each layer to prevent overfitting. The output layer employs a Softmax activation function for multiclass classification. Table 1 outlines the detailed parameters of the proposed WST-CNN model architecture. The model has a total of 2.18 M trainable and 1.44 K nontrainable parameters, respectively.

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IV. EXPERIMENTAL METHODOLOGY

A. Datasets

1) BCI Competition IV-2a [13]: The BCI Competition IV-2a [13] dataset comprises EEG data collected from nine participants, each performing four distinct MI tasks: left-hand (LH), right-hand (RH), feet (FT), and tongue (T). The data was gathered using 22 EEG channels, with signals bandpass filtered between 0.5 Hz and 100 Hz, and sampled at a rate of 250 Hz according to the standard international 10–20 electrode setup. Each participant's data includes two sessions—one for training and another for testing—captured on different days. Each session consists of six runs, with a short interval between each run. Each run consists of 48 trials, with 12 trails per each class. The duration of a trial was 7.5 s.

2) DBCIE [14]: The DBCIE [14] dataset comprises EEG data collected from 23 healthy individuals (17 males and 6 females, mean age = 27 years, SD = 4). Each EEG recording is 180 s long. The data were recorded at a sampling rate of 256 Hz with a 16-bit resolution and bandpass filtered between 0.5 Hz and 30 Hz. EEG signals are gathered via three electrode placement methods—Montage 1, Montage 2, and Montage 3—following the international standard protocol. Recordings are made in three different MI activities for each participant: left-arm (LA) movement, right-arm (RA) movement, and both-arms (BA) movement. For this analysis, we considered Montage 3 placement method which has 32 channels, (refer [14]). Each participant's EEG data was captured as a single ".csv" file.

B. Experiment Setup

1) Preprocessing: The acquired EEG signals are first filtered to remove noise and different artifacts. A fifth-order Butterworth bandpass filter with cut-off frequencies 0.5–30 Hz has been employed to remove noise and artifacts from recorded EEG data. Thereafter, each signal is fragmented into nonoverlapping segments of 4 s. The segmented EEG data is fed into a WST layer followed by 1-D-CNN.

2) Model Details: Incorporating the second-order WST layer employs a Morlet wavelet with $Q = 12$, and the average scale is set to 2^J, where $J = 8$. In the convolutional blocks, the first convolutional layer consists of 256 filters with a kernel size of 5, followed by a dropout of 0.3. The second convolutional layer employs 128 filters with a kernel size of 3, followed by a dropout rate of 0.3. LeakyReLU has been employed as the activation function. Following the convolutional block, two affine layers (AL1, AL2) with 196 and 128 neurons, followed by dropout rates of 0.3, respectively, are used for classification. Finally, at the output layer, the softmax activation function was used to achieve multiclass classification. The number of epochs, learning rate, and batch size have been set to 200, 0.0001, and 128, respectively. Adam and categorical cross entropy were chosen as the optimizer and the loss function, respectively.

3) Training Paradigm: To produce robust and generalized analysis results, a fivefold cross-validation (CV) technique has been used for training and test sets. The samples were arbitrarily divided into five subsets (K1 → K5), and each of which maintains the proportionate distribution of categories in the dataset. The proposed model was trained on four subsets, and the remaining subset holds out for testing. The training and testing sets are split in the ratio of 80:20, while 15% of training samples were used as validation samples during the training of the model. The training (and subsequent test) performance is evaluated based on accuracy (η), sensitivity (Se), specificity (Sp), and F1 score.

4) Computational Details: The experiments are conducted using a GPU Tesla P100-PCIE-16GB, RAM-16GB, TensorFlow-Keras version 2.15.0, and Python 3.10. The proposed method's training time is 13 218.68 s (46.94 s for testing) for the BCI Competition IV-2a dataset and 22 517.41 s (78.39 s for testing) for the DBCIE dataset.

Table 2. Performance Metrics (%) for All Classes on BCI Competition IV-2a [13] and DBCIE [14] Datasets, Respectively

5-fold CV	Performance Metrics (%)							
	BCI Competition IV-2a [13] dataset				DBCIE [14] dataset			
	η	*Se*	*Sp*	*F1-Score*	η	*Se*	*Sp*	*F1-Score*
K1	92.50	92.50	97.50	92.75	89.19	89.19	94.60	89.33
K2	92.45	92.45	97.48	92.50	88.88	88.88	94.44	89.00
K3	92.72	92.72	97.57	92.75	89.00	89.00	94.49	89.00
K4	91.66	91.66	97.22	92.00	88.91	88.91	94.45	89.00
K5	91.90	91.90	97.30	92.00	89.05	89.05	94.52	89.00

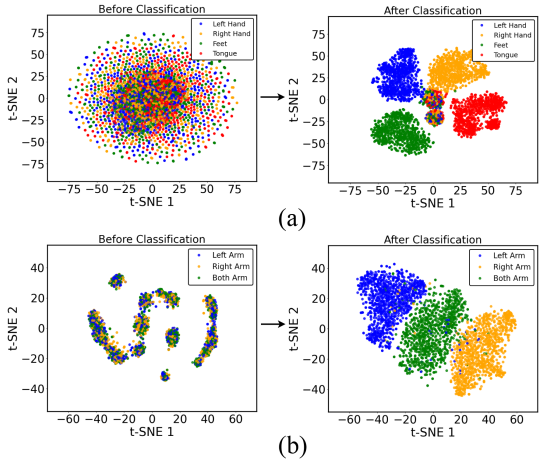


Fig. 2. Visualization of the feature distributions using (a) BCI competition IV-2a [13] and (b) DBCIE [14] datasets, respectively.

V. RESULTS AND DISCUSSION

This section discusses the performance of the proposed WST-CNN model for MI task classification. In addition, comparisons with SOTA methods are also explored.

A. Discussion

Table 2 provides a summary of the fivefold CV findings for all classes on both datasets, BCI Competition IV-2a [13] and DBCIE [14], respectively. In Table 2, the first column represents the different folds followed by the performance metrics (%)—η, S_e, S_p, and F 1-scores, on both datasets, respectively. The classification measures η, S_e, S_p, and F 1-score for the BCI Competition IV-2a dataset reach up to 92.72%, 92.72%, 97.57%, and 92.75%, respectively. In a similar fashion, the DBCIE dataset yields an F1-score of 89.33%, an η and S_e of 89.19%, and a S_p of 94.60%. Based on these findings, it can be concluded that the model is good at recognizing true negative cases because it performs well in differentiating between classes, with very high specificity in both datasets.

Fig. 2 illustrates the T-distributed stochastic neighbor embedding (t-SNE) [22] visualization of feature distributions from both datasets before and after classification using the proposed method. It can be seen that the significance of the WST-CNN in extracting the distinctive features from EEG data for each class, resulting in efficient clustering.

B. Ablation Study

It is crucial to know whether any specific type of MI has an impact on the overall classification performance. To this end, different combinations of MI tasks for the BCI Competition IV-2a dataset have been analyzed, with the results summarized in Table 3, indicating the proposed method efficiently classifies MI tasks from EEG. We examined how various combinations of MI tasks affect categorization performance. Each row represents a unique combination, and by

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Table 3. Classification Performance Recorded With Different Combinations of Three MI Tasks (BCI Competition IV-2a [13] Dataset)

MI Tasks			Performance Metrics (%)			
			η	S_e	S_p	$F1-Score$
Left Hand	Feet	Tongue	93.21	93.24	96.62	93.31
Left Hand	Right Hand	Feet	93.09	93.11	96.55	93.19
Left Hand	Right Hand	Tongue	93.31	93.32	96.66	93.40
Right Hand	Feet	Tongue	93.24	93.31	96.65	93.33

Table 4. Comparative Study With Related Works

Methods	MI Tasks	Performance Metrics (%)			
		η	Se	Sp	$F1-Score$
BCI Competition IV-2a dataset [13]					
SPCNN [17]	4	71.80	71.93	90.61	71.62
CNN1D_MFBF [16]	4	73.17	73.17	88.17	73.10
ATCNet [18]	**4**	**85.05**	**85.05**	**95.01**	**84.72**
WST-CNN (Proposed)	**4**	**92.25**	**92.25**	**97.41**	**92.40**
DBCIE dataset [14]					
ATCNet [18]	**3**	72.08	72.17	86.04	72.03
CNN1D_MFBF [16]	**3**	**80.91**	**80.92**	**90.46**	**80.95**
SPCNN [17]	**3**	**84.22**	**84.01**	**93.64**	**84.00**
WST-CNN (Proposed)	**3**	**89.01**	**89.01**	**94.50**	**89.07**

comparing them, we may identify which tasks contributed the most to overall performance. The first column represents the combinations of different MI tasks followed by the performance metrics (%). As seen, the proposed approach achieved the maximum classification η, S_e, S_p, and F 1-score for the combination of LH, RH, and T (third row), i.e., FT may have some similar characteristics with other MI tasks, resulting in lower performance when it is included. Conversely, the minimum classification performance was noted for the combination of LH, RH, and FT (second row) without T. Therefore, during the discrimination of MI tasks, FT and T MI tasks should be carefully handled. Despite these variations, all the classification results are very consistent, validating the efficacy of the proposed model.

For comparison purposes, the following SOTA methods are presented: CNN1-D_MFBF [16], SPCNN [17], and ATCNet [18]. Table 4 presents the performance metrics (%) of the proposed method compared to other SOTA methods. Using fivefold CV, the proposed method achieved a mean accuracy of 92.25% with a standard deviation (SD) of 0.40% (the lowest among others, 73.17%, 71.80%, and 85.05%, respectively, for CNN1-D_MFBF, SPCNN, and ATCNet), for MI classifying tasks on BCI Competition IV-2a dataset. For the DBCIE dataset, the proposed method obtained a mean accuracy of 89.01% with a SD of 0.11% (in comparison to 80.91%, 84.22%, 72.08%, for CNN1-D_MFBF, SPCNN, and ATCNet, respectively), for LA, RA, and BA, respectively. The proposed method observes the best confidence intervals (CIs) of [91.90%, 92.60%] on the BCI Competition IV-2a dataset and [88.91%, 89.10%] on the DBCIE dataset. The second-best CIs are [83.62%, 86.48%] for ATCNet [18] in the BCI Competition IV-2a and [80.83%, 81.00%] for CNN1-D_MFBF [16] in the DBCIE. These results highlight the statistical significance of the proposed method in classifying MI tasks.

VI. CONCLUSION

This study proposes a novel approach for classifying MI tasks using EEG data via WST-CNN. WST effectively captures robust features from EEG signals that are invariant to translations and small deformations, reducing noise and variability. WST reduces the computational burden on CNNs. Furthermore, we have studied different combinations of three MI tasks of the BCI Competition IV-2a dataset. Such a novel study may prove to be critical in seizure analysis where neuromotor signals of multiple limbs are involved. In the experiment, the proposed concept was validated on the BCI Competition IV-2a dataset and the DBCIE dataset, respectively. As can be observed, the proposed approach outperforms the SOTA techniques. The proposed concept can effectively differentiate MI activities using EEG data, demonstrating favorable experimental performance. In the future, we will explore the use of various WTs and assess their impact on MI task classification. The demo code of the proposed method can be found online.

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Data is available online at https://github.com/Rantu1027/WST_CNN.

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